

Sampling Distributions, Addendum

CSU, Chico Math 351

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Sampling Distributions Reminder

All statistics have a sampling distributions. The sample mean is relatively easy to calculate, and thus Statistics (the discipline) most often estimates population means.

Sample Median

Assume we have $X_1, \dots, X_n \sim_{iid} F$, with probability density function f . Define m such that $P(X \leq m) = 0.5$. It can be shown that the sampling distribution of the sample median $\tilde{X}$ is approximately normal when the sample size is sufficiently large,

$$\frac{\tilde{X} - m}{4nf^2(m)} \dot{\sim} N(0, 1). \quad (1)$$

Sample Median, notes

Note that we want a statement like Equation (1) where we don't have to assume we know the distribution F or the density function f – the Central Limit Theorem gives us this for the sample mean.

How do they compare?

If sample mean and sample median are both unbiased (for the Normal distribution), how do their variances compare?

As seen earlier, for large sample size N , the sample median follows an approximate Normal distribution. For the normal distribution, the variance will be $\frac{\pi\sigma^2}{2N}$ for the median, versus $\frac{\sigma^2}{N}$ for the mean. Thus the mean will still have lower variance in comparison.

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Sample Median

References

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Michael G. Akritas. *Probability and Statistics with R for Engineers and Scientists*. Pearson Education, Inc., 2016.

Adapted from notes by Dr. Roualdes.